Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[usenames,dvipsnames,10pt,aspectratio=169]beamer
[utf8]inputenc verbatim []beamerthemeims
[style=authoryear,backend=biber]biblatex bibliography.bib
authorgiven-family
silence biblatexPatching footnotes failed

On The Explosion of Large Death Stars

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IMS Innovation Group

19 Pebruari 2025

Daftar Isi

Before you start

Colors

Blocks

Enumerates, itemizes and description

- Enumerates and itemizes
- Description

Maths

Other blocks

Bibliography and Publications

Bonus Commands

Overleaf users

Warning

You can ignore this slide if you're **not** working with Overleaf.

Overleaf, Beamer and Biber do not always get along well together. For this reason, if you make a mistake while writing this presentation, in the drop-down error message you'll **always** get Biber-related error messages.

Luckily, you just have to click on "`go to first error/warning`" and the UI will scroll to the line containing your mistake.

Compiling

Warning

You can ignore this slide if you **are** working with Overleaf.

To compile this deck you'll need the `biber` package. Probably your TEXeditor already supports it; if not, you will easily find online the instructions to install it.

If you're not using an editor, you can compile this presentation using the command line by running:

```
$ pdflatex main.tex
$ biber main.bcf
$ pdflatex main.tex
$ pdflatex main.tex
```

Colors

For this template we defined two colors:

-
-

You can use these colors as you want in your presentation. For example, you can **color the text in orange** by writing `\textcolor{IMSOrange}{my green text}`.

We also redefined many of the most common LaTeX and Beamer commands, like `itemize`, `block`, etc. You will see samples of these commands in the following slides.

This is a page with a title and a subtitle

And also some blocks.

Goal of the mission

Shoot in the Death Star's exhaust port and destroy it before it can fire on the Rebel base.

Take care!

TIE Fighters may chase you while approaching the target.

Use the force you must

Remember your training with Obi-Wan, and use the Force to make the perfect shot.

Enumerates and itemizes

This is an example of `itemize`.

- A long time ago in a galaxy far, far away...

And this is an example of `enumerate`.

1. Go to the Death Star.
2. Find the exhaust port.
3. Make the perfect shot.
4. Become a hero.

Description

This is an example of `description`.

Luke No. No! That's not true! **That's impossible!**

Description

This is an example of `description`.

Vader *I am* your father.

Luke No. No! That's not true! **That's impossible!**

Description

This is an example of `description`.

Vader *I am* your father.

Luke No. No! That's not true! **That's impossible!**

And while we're here, let's have a look to `verbatim` as well, to see how we made items appear in arbitrary order:

```
\begin{description}
  \item<2->[This is the first item - appears after] one
  \item<1->[This is the second item - appears first] two
\end{description}
```

Maths

A formula will look like this:

$$x^2 + y^2 = z^2$$

You can number equations as well:

$$1 + 1 = 2 \tag{1}$$

$$1 + 1 = 2 \qquad \text{(custom label!)}$$

If you want to use the sans serif math fonts, or use a serif font for the main text, just go to `beamerfontthemeims.sty` and select the indicated font option.

Theorems

The usual `theorem`, `corollary`, `definition`, `definitions`, `fact`, `example` and `examples` blocks are available as well.

Theorem

There exists an infinite set.

Proof.

This follows from the axiom of infinity. □

Example (Natural Numbers)

The set of natural numbers is infinite.

Other blocks

Here we display examples of `abstract`, `verse`, `quotation`, and `quote`.

Ringkasan

This is an abstract.

This is a verse.

> *This is a quotation.*
>
> *-Han Solo*

> *A quote this is.*
>
> *-Yoda*

Bibliography

You can cite an article

- normally using `\cite`, e.g.: ([?])
- or display the full citation using `\fullcite`, e.g.: article1

  (n.d.) stands for "no date". `year={A long time ago...}` is not a date that should be specified in `bibliography` anyway.

Look at the code of the following slide to see how to automatically split the bibliography on many slides. You can also use `\nocite{*}` to display the non-cited publications as well.

Bibliography I

Framecard

You can display a frame with a colored background and a huge text in the center using the command `\framecard`.

For example, you can write:

```
\framecard{A SECTION\\TITLE}
```

This will display a frame with a blue background and the phrase "A SECTION TITLE" in the center. You can also use a custom color with `\framecard`:

```
\framecard{A SECTION TITLE}
\framecard[IMSOrange]{A SECTION TITLE\\
WITH A CUSTOM COLOR}
```

You can see the results of the commands above in the following slides.

A SECTION TITLE [IMSOrange]A SECTION TITLE WITH A CUSTOM COLOR

Framepic

You can display a frame with a background image using the command `\framepic`. The image will be **adapted vertically** to fit the the frame.
For example, you can write:

```
\framepic{graphics/darth}{
 \framefill
    \textcolor{white}{Luke,\\I am your supervisor}
    \vskip 0.5cm
}
```

Alternatively, to make the background 50% transparent, you can write `\framepic[0.5]{graphics/darth}...`
You can see the results of the commands above in the following slides.

graphics/darth

[0.5]graphics/darth

Right-aligned text with
Semi-transparent background

Other bonus commands I

We provide two other bonus commands:

`pdfnewline` you can use `\pdfnewline` to avoid the annoying `hyperref` related warnings when using newlines in the document's title, author, etc. For example, in this presentation the author is defined as:

```
\author[Luke Skywalker]{
  Luke Skywalker, Ph.D.
  \pdfnewline
  \texttt{luke.skywalker@umu.se}
}
```

`marker` you can use `\marker` to highlight some text. The default color is orange, but you can also [IMSBlue]use a custom color. For example:

```
\marker{Default color}
\marker[IMSBlue]{Custom Color}
```

Other bonus commands II

`framefill` you can use `\framefill` to put the text at the bottom of a slide by filling all the vertical space.